

Introduction:

In the world of tennis, Wimbledon is widely regarded as one of the most prestigious tournaments in the world, where all of the top male and female players gather to compete for the coveted title. To be successful in tennis, winning important points is crucial and can change the outcome of a game. For example, during a break point opportunity, meaning when one player has the opportunity to win a game when the other player is serving, winning the point can drastically change the momentum and course of the game. Using a Markov chain model, I aimed to create a metric analyzing who the top male and female players were in the 2023 Wimbledon tournament based on win probability added over the course of all of their respective matches in the tournament.

State Space:

The tennis play by play dataset that I used provided a multitude of details about each specific point, but to help with computational efficiency and to ensure the state space was not too granular, I only included a few of these details in the state space. The state space includes which player is serving, how many games the first player has won in the set they were playing in, how many games the second player has won in the set they were playing in, and how many sets the first and second player had won so far. Initially, I also had put the current score of the game the players were playing in(ex. 15-40), but I waited a significant amount of time for the value iteration to run without it finishing, so I decided to remove the game score to significantly minimize the state space.

Leaderboard:

The dataset I used contained match information for both male and female players, so the leaderboard of the top 10 players by win probability added contain both genders, although I will separate by gender later on. The leaderboard is as follows:

Rank	Player	Win Probability/Wins Added	Number of Matches Played
1	Elina Svitolina(F)	7.35	6
2	Novak Djokovic(M)	6.44	7
3	Marketa Vondrousova(F)	5.82	7
4	Holger Rune(M)	4.85	5

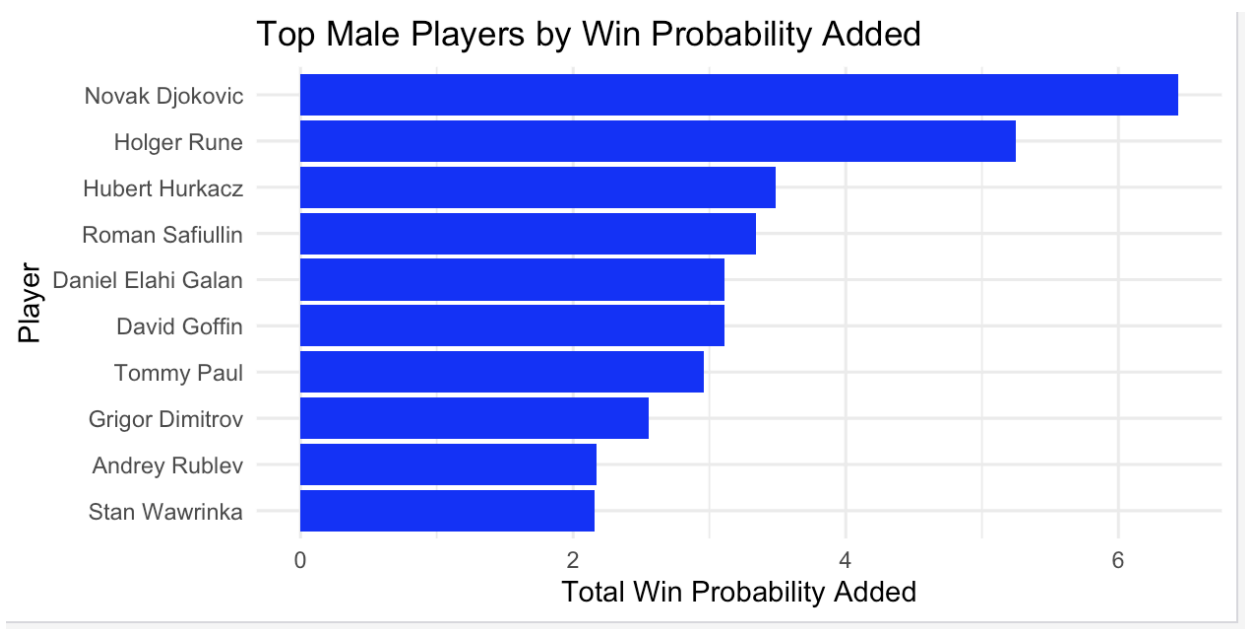
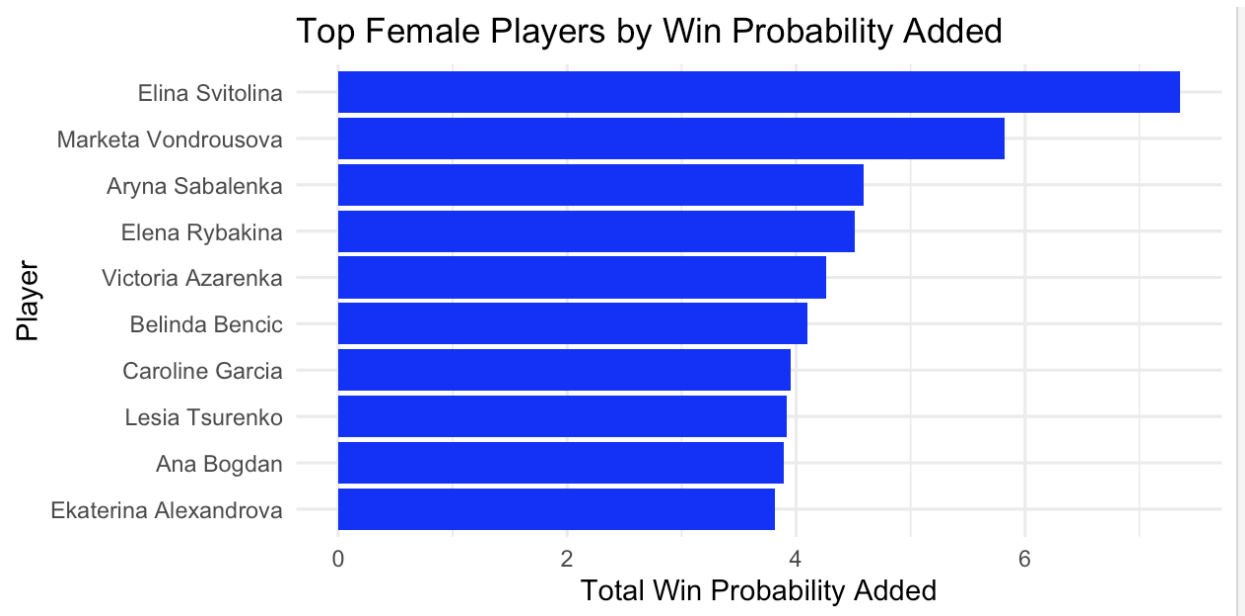
5	Aryna Sabalenka(F)	4.59	6
6	Elena Rybakina(F)	4.51	5
7	Victoria Azarenka(F)	4.27	4
8	Belinda Bencic(F)	4.10	4
9	Caroline Garcia(F)	3.95	3
10	Lesia Tsurenko(F)	3.92	4

From this leaderboard, one trend that is immediately clear is the effect that more matches played will have on the metric, which makes sense. Playing more matches gives the players more opportunities to win key points, subsequently increasing their total win probability added. However, one flaw and error in the results is that players such as #1 ranked Elina Svitolina has a win probability added that is greater than the total number of matches played. One reason for this error could be that female players only play up to 2 sets, while male players play up to 3 sets. As a result, with fewer sets, games, and points required to win a match, the model may value female players' actions and results significantly more than males, as there are less states needed to win. With that said, Svitolina's extremely high win probability added generally makes sense for the most part, as she went on a run to the Wimbledon semifinals, defeating the world #1 ranked female player, Iga Swiatek, in the process. At the time, Svitolina was ranked around 80th in the world, and her run to the semifinals is consistent with the metric indicating that she must have won many crucial points and games in the process to do so.

For the males, Novak Djokovic has the highest win probability added with 6.44 wins added over the course of 7 matches played. Djokovic is widely regarded as one of the greatest tennis players of all time, and as a player with veteran experience who had previously won 7 Wimbledon championships, it makes sense that he is atop the male leaderboard. Additionally, in this specific year, Djokovic played 7 matches and made it to the finals, so his high win probability added is also a result of playing many matches and having more opportunities to win points than most other players in the tournament. Holger Rune was ranked second for the males with 4.85 wins added over 5 matches, where he lost in the quarterfinals to eventual champion Carlos Alcaraz. In these 5 matches, Rune played 6 total sets where he won 7 games to 6, which is the maximum number of games that could be played in a set where there is also a tiebreak played

for the last game to determine who ultimately wins the set. This is an abnormally high amount of 7-6 sets played in such few games, but it explains why Rune's win probability added is so high. Winning these close sets and tiebreakers is crucial to a player's chances of winning the match, and Rune won 5 out of the 6 7-6 set opportunities he had, demonstrating that he performed well in pressure situations throughout the tournament.

Separated Male and Female Leaderboard:



Above are bar charts displaying the top 10 female and male players by win probability added, and the charts above further illustrate the discrepancy between genders with this metric. Other

than Djokovic, most of the women clearly have higher win probabilities added, which is a trend that continues beyond the top 10. Again, it can likely be attributed to the fact that the female players play less sets than the males, meaning that each of their actions, especially in crucial moments, will have a more drastic effect on the win probability. However, one point I wanted to bring up is the exclusion of Carlos Alcaraz from the top 10 for male players in win probability added. Alcaraz was the ultimate champion of Wimbledon in 2023 and one of the main motivations I wanted to look at this tournament in particular. Alcaraz had a thrilling 5 set match against Novak Djokovic in the final, and this tournament ultimately rocketed Alcaraz into fame, where he is widely regarded as the best current male tennis player in the world. Surprisingly, Alcaraz is ranked nearly last in win probability added, with him having -0.963 wins added over 7 matches played. This is extremely surprising, as considering that Alcaraz won the championship, it would be assumed that his win probability added would be somewhat high and he would have needed to win many crucial points. One possible explanation for this could be that Alcaraz had a relatively dominant run to the championship in 2023. In the 7 matches that he played, Alcaraz won 4 of them in 3 sets, the fewest needed to win. Additionally, he only played 4 sets in those 7 games that was a 7-6 set, meaning that Alcaraz generally won in dominant fashion and did not face many crucial points. With that being said, the final Alcaraz played in against Djokovic had many pressure points involved in it, but it was likely not enough to outweigh the blowout games and sets that Alcaraz played in matches prior.

Conclusion:

Ultimately, I thought that using a Markov Chain to create a win probability/wins added metric was helpful and applicable to the sport of tennis, as there are many crucial moments that could swing the outcome of a match. However, I would have liked to include more factors, such as the game score, in the state space to make the metric more accurate and inclusive, and I would have liked to find a way to do so without making the model too computationally expensive.

Additionally, using a Markov Chain to evaluate female players as a collective and male players as a collective would be effective, but I had both female and male players in my dataset, which potentially led to overvaluing of female players in my metric.